GAUTAM SINGH

Software Engineer — backend, data, and ML systems

Guelph, ON | (226) 979-6678 | gautamsingh2189@gmail.com | linkedin.com/in/gautam-singh-cs | github.com/Gautamstar | gautamstar.github.io/portfolio

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Rust, Java, C++
Backend & Frontend: FastAPI, Django, Node.js, React, REST APIs, Redis/RQ, Flutter
Cloud & DevOps: AWS (Lambda, DynamoDB, CloudFront), Azure, Docker, Terraform, Linux, GitHub Actions CI/CD, Cloudflare
Data & ML: PostgreSQL, MongoDB, Redis, Databricks, PySpark, PyTorch, scikit-learn, Pandas, OpenCV, YOLO

EXPERIENCE

Machine Learning Intern Aug 2026 - Present
Sphere Global (automated vehicle damage inspection) *Remote*

- Build and maintain the Python tool the labeling team uses every day to prepare training data for an automated vehicle damage inspection system, running on one ordinary laptop with no GPU so customer photos never leave the site.
- Created a generator that turns clean vehicle photos into ~12,000 realistic damaged ones in NumPy and OpenCV, each labeled automatically to the pixel in YOLO format, removing the need to find, photograph, and hand-mark damaged cars.
- Benchmarked SDXL and FLUX, two leading AI image generators, and showed the in-house generator reproduced real damage more accurately, faster, and at lower cost, which kept the team off a costly dead end.
- Found and fixed a flaw that had mislabeled 39% of training images, where the tool outlined the floor and walls instead of the car, by making a YOLO detector confirm the vehicle before SAM 2.1 traced it.

ServiceNow Developer & QA Specialist Sep 2024 - Dec 2024, Sep 2025 - Dec 2025
PAMT Consulting (now Ateko, a Bell Canada company) *Remote*

- Reduced new-hire onboarding handling time by ~30% by replacing a manual, error-prone process with an automated workflow built in ServiceNow App Engine Studio and Flow Designer.
- Designed and ran test cases across development, test, and production environments before each release.

PROJECTS

Distill *2026*
Python, PyTorch, Qwen2.5-VL, Transformers, Pydantic, FastAPI, Docker, GitHub Actions

- Built a local document extraction service that reads invoices and receipts into typed JSON and scores its own confidence on every field, lifting delivered accuracy from 80% to 97% by sending only the fields it does not trust to a human.
- Kept the published numbers honest with 175 tests and a CI gate that fails the build when a metric moves.

FitFileSize Live | GitHub | *2026*
Python, FastAPI, Redis/RQ, React, TypeScript, Docker, Cloudflare Tunnel, Ghostscript, MCP

- Solo-built and launched a web app, public API, and MCP server that compress PDFs and images (7 formats, including iPhone HEIC) to fit an upload limit or a form's exact rules, such as visa photos and exam uploads.
- Designed an async job pipeline (FastAPI, Redis/RQ workers, Server-Sent Events) with format-specific logic behind one strategy interface, self-hosted with Docker on a VPS behind a Cloudflare Tunnel with deploy-on-merge.
- Halved the renders per compression with a guided search that predicts the right setting from sizes already measured, taking a 5 MB photo from 9.2 s to 4.2 s on production with byte-identical output, behind ~350 tests across three CI jobs.
- Hardened the deployment from failure drills, so a watchdog restarts any container the health endpoint reports hung and a deploy gives workers two minutes to finish where Docker allowed ten seconds.

Cairn Live | GitHub | *2026*
Rust, AWS Lambda, DynamoDB, CloudFront, API Gateway, Terraform

- Built the ingest path as a Rust binary on ARM64 AWS Lambda that responds in 5.1 ms warm and 184 ms cold, doing one DynamoDB write per event and deferring every dimension derivation.
- Deployed the full stack of Lambda, DynamoDB, CloudFront, and API Gateway as 41 Terraform managed resources running live across three sites at zero steady-state cost and made the service cookie-free by hashing visitor identifiers under a key that rotates daily.

The Almanac (Capstone) GitHub | *2025*
Python, Azure Data Factory, Databricks, PySpark, Azure SQL, PyTorch, Django REST, Flutter

- Redesigned the storage layer of an automated market-data pipeline behind a stock-tracking web and mobile app, improving data load performance by ~35%; shipped with a team over eight months.

OPEN SOURCE

actualbudget/actual PR #8603 | *Merged 2026*

- Contributed a merged bug fix to a widely used open-source personal finance app, returning a proper error when a password change is attempted with no password method set.

EDUCATION

Bachelor of Computer Science (Honors) Sep 2020 - Dec 2025
Conestoga College *Waterloo, ON*

Google Advanced Data Analytics Professional Certificate *Jun 2026*